# PROPER PUSHFORWARD OF CYCLE AND CHOW GROUPS COMMUTES WITH THE BASE CHANGE PULLBACK IN THE SHIFTED DEGREES

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

*Editorial note.* The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

*Editorial note.* Canonical location of the counted proposition: Stacks tag 0FVL (sources/stacks-chow/pinned-original-source.tex line 15340).

*Allowed standard prerequisites (editorial).*

- The base change situation with the constant shift c and the function delta'. Same chapter, Situation labelled situation-setup-base-change (source line 15123), invoked in the statement. Fixes the shift c and the dimension conventions in which the two diagrams are stated.
- The cycle attached to the pushforward of a closed subscheme's structure sheaf. Same chapter, Lemma labelled lemma-cycle-push-sheaf (source line 1992), invoked at proof lines 15366-15368. Supplies the identifications of f_*[Z] and (f')_*g^*[Z] with the fundamental cycles of the pushed forward structure sheaves.
- Base change of the k-cycle of a coherent sheaf. Same chapter, Lemma labelled lemma-pullback-coherent-base-change (source line 15193), invoked at proof lines 15369-15370. Supplies g^*f_*[Z] = [(X' -> X)^* f_*O_Z]_{k+c}; it is counted separately in this lane as S12-algebra-pullback-coherent-base-change.
- Flat base change for the cohomology of a quasi-coherent sheaf. Cohomology of Schemes, Lemma labelled coherent-lemma-flat-base-change-cohomology (Stacks tag 02KH), invoked at proof lines 15375-15378. Supplies the module isomorphism that closes the argument.

*Source reference.* Original reference 0FVG: sources/stacks-chow/pinned-original-source.tex, source lines 15123–15123. Same-chapter pointer into the author environment situation at source line 15123; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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*Date*: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Source reference.* Original lemma 02R6: sources/stacks-chow/pinned-original-source.tex, source lines 1992–1992. Same-chapter pointer into the author environment lemma at source line 1992; invoked at proof line 15366 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 0FVI: sources/stacks-chow/pinned-original-source.tex, source lines 15193–15193. Same-chapter pointer into the author environment lemma at source line 15193; invoked at proof line 15369 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

*Source reference.* Original lemma 02KH: sources/stacks-chow/tags.txt, source lines 3372–3372. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 15378 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

**Lemma 0.1.** *In Situation 0FVG let $Y \to X \to S$ be locally of finite type and let $Y' \to X' \to S'$ be the base change by $S' \to S$. Assume $f : Y \to X$ is proper. Then $f' : Y' \to X'$ is proper and the diagram*

$$\begin{array}{ccc} Z_k(Y) & \xrightarrow{g^*} & Z_{k+c}(Y') \\ {\scriptstyle f_*}\downarrow & & \downarrow{\scriptstyle f'_*} \\ Z_k(X) & \xrightarrow{g^*} & Z_{k+c}(X') \end{array} \quad \text{and} \quad \begin{array}{ccc} \mathrm{CH}_k(Y) & \xrightarrow{g^*} & \mathrm{CH}_{k+c}(Y') \\ {\scriptstyle f_*}\downarrow & & \downarrow{\scriptstyle f'_*} \\ \mathrm{CH}_k(X) & \xrightarrow{g^*} & \mathrm{CH}_{k+c}(X') \end{array}$$

*of cycle and chow groups commutes.*

*Proof.* It suffices to show the first diagram commutes. To see this, let $Z \subset Y$ be an integral closed subscheme of $\delta$-dimension $k$ and denote $Z' \subset X'$ its base change. By construction we have $g^*[Z] = [Z']_{k+c}$. By Lemma 02R6 we have $(f')_* g^*[Z] = [f'_* \mathcal{O}_{Z'}]_{k+c}$. By the same lemma we have $f_*[Z] = [f_* \mathcal{O}_Z]_k$. By Lemma 0FVI we have $g^* f_*[Z] = [(X' \to X)^* f_* \mathcal{O}_Z]_{k+r}$. Thus it suffices to show that

$$(X' \to X)^* f_* \mathcal{O}_Z \cong f'_* \mathcal{O}_{Z'}$$

as coherent modules on $X'$. As $X' \to X$ is flat and as $\mathcal{O}_{Z'} = (Y' \to Y)^* \mathcal{O}_Z$, this follows from flat base change, see Cohomology of Schemes, Lemma 02KH. $\square$